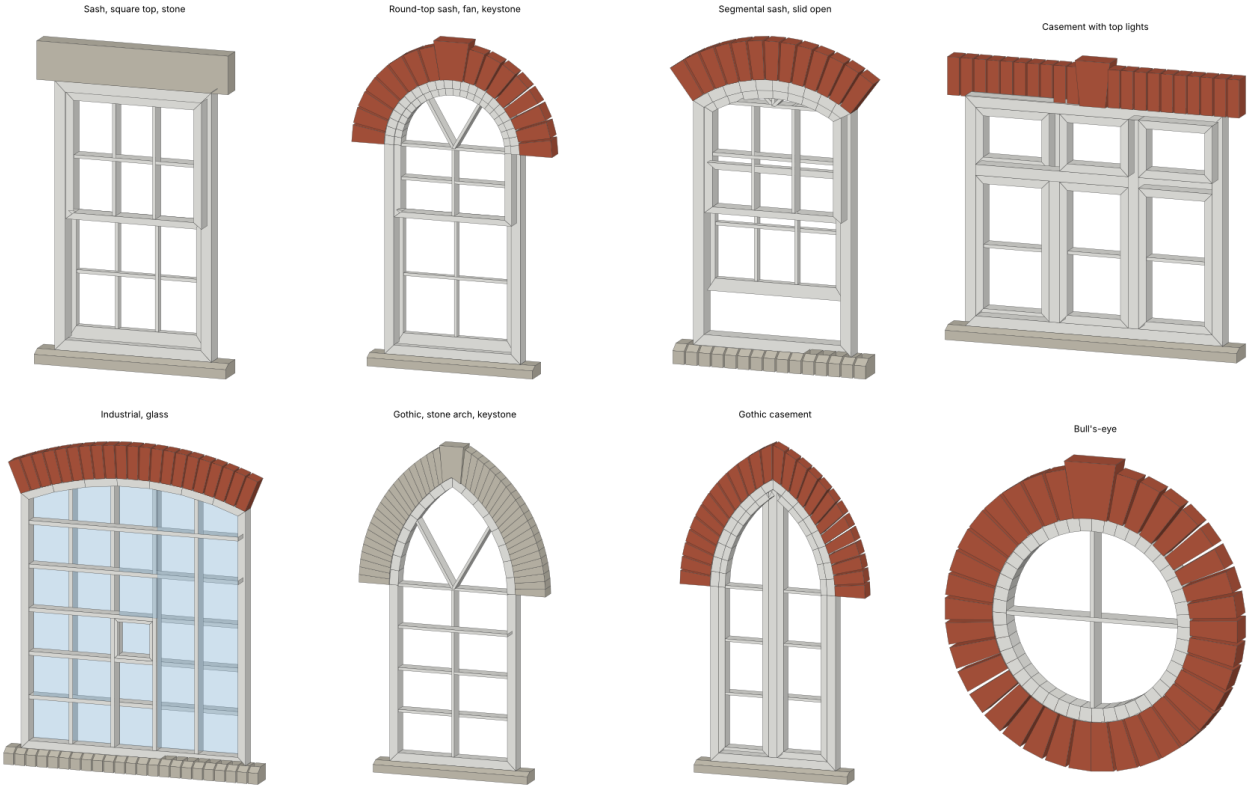
3D Period Windows

User guide - plugin for IngeTrazo (version 9)



What it does

3D Period Windows makes old-fashioned windows that fit the face you select. Draw the shape of a window on a wall (or a sloping roof), select it, and the plugin builds a complete window to that outline: frame, sashes or casements, glazing bars, optional glass, a stone or brick sill, and a stone or brick head with an optional keystone. It recognises square, round-topped, segmental, pointed (gothic) and round (bull's-eye) shapes, and can cut the opening right through a solid wall so the window, sill and head all sit flush with the back of the wall - ready to export and 3D print flat, with a clean wall face left for brickwork.

- **Three styles:** sliding sash, side-hung casement (with mullions, transom and top lights), and fixed / industrial (a grid of bars with an optional pivoting pane).
- **Panes in columns and rows**, set separately for the top and bottom sash (6 over 6, 2 over 2 ...), with fan bars, a carried-on grid or plain glass in an arched top.
- **Sills and heads:** sloped stone sill or brick-on-edge sill; stone lintel or stone arch; brick soldier course or brick arch; tapered keystone.
- **Made for printing:** everything flat on one side, each window joined into one watertight solid, and a hair's gap to the wall so slicers don't get confused.
- **Presets** for common windows, and your own presets saved as files.
- **Live preview** in 3D, and a lettered **sizes drawing** (A, B, C ...) that matches the size boxes.

1. Installing and opening

1. Copy **period_windows.py** into your IngeTrazo plugins folder (the same folder as your other 3D plugins).
2. Close IngeTrazo completely and start it again.
3. Open the plugin with **Ctrl+Alt+N**, from the **Extensions** menu, or with the **Windows** button on the 3D Plugins toolbar (if you have plugins_toolbar.py installed).
4. Check the window title says **3D Period Windows (version 9)**. If it shows an older number, IngeTrazo is still loading an old copy of the file.

The window stays open while you work: select a face in the model, click **Add window to selected face(s)**, select the next face, add again, and so on. Press Ctrl+Alt+N while it is open to bring it to the front.

2. Quick start: a window in a solid wall

1. Make the wall as a solid (a box with a front and a back face).
2. Draw the window shape on the front face: a rectangle, a rectangle with a round or arched top, a pointed arch or a circle. IngeTrazo splits the wall face so the shape becomes its own face.
3. Select that shape (you can select several shapes and do them all at once).
4. In the plugin pick a **Preset**, or choose the **Window style** and **Top shape** yourself. *Follow the face shape* uses the outline you drew.
5. Set the panes, sill and head on the other tabs. The preview updates as you go.
6. Click **Add window to selected face(s)**.
7. Read the **Last window added** message at the top of the plugin window (see section 8).
8. When all the windows are in, select the wall's front face and run **3D Brick & Block Studio** for the brickwork. The bricks stop a hair short of every opening.

Everything one click of **Add window** does - the new window groups, the removed face and the cut through the wall - is one undo step. Ctrl+Z puts the wall back as it was.

3. Which face to select

You select	What happens
A shape drawn on a **solid wall**	The window fills the shape. The shape is removed and the window, sill and head are cut right through the wall and sit flush with its back face.
A shape drawn on a **single sheet** (no back face within 2 m)	The window replaces the shape and sits on the face, a hair (the gap) in front of it.
A **wall face that already has openings**	A window goes into every opening. The wall face is kept. The window fills the opening from the back of the wall.
Several faces	Each one gets its own window, all in one go.
A **sloping roof** face	Works the same: 'up' runs up the slope, so the sill is at the lower edge and the head at the upper edge (set them to None for a rooflight).
A **flat roof or floor** face	The window lines up with the face's longest edge.

If the sill and head come out on the inside of the wall, tick **Window faces the other way** and add it again (cutting through the wall is skipped when this is ticked).

4. The Window tab

Setting	What it does
Window style	**Sash** (two sliding sashes with a meeting rail), **Casement** (hinged leaves) or **Fixed / industrial** (one grid of glazing bars).
Top shape	**Follow the face shape** (recognises square, round, segmental, pointed and round outlines; any other shape is used as it is, with a flat head), **Square top**, **Round top (semicircle)**, **Segmental**, **Pointed (gothic)** or **Round (bull's-eye)**.
L Arch rise	How high a segmental arch rises above its springing.
Flat back for 3D printing	On by default. Every part goes right down to the back, so the window prints flat. Casements and pivot lights can't be shown open like this; sliding sashes can.
Join into one solid	On by default. Uses IngeTrazo's own Union so the window, sill and head come out as one watertight solid. Glass stays as its own group.
Remove the face	The face the window is placed on is taken away (a face with openings is kept).
Cut through the wall	When the face is on a solid wall, the window, sill and head are cut right through it and sit flush with the back of the wall.
Gap round the window	A tiny gap (0.01 mm) between the window and the wall so slicers don't see faces lying exactly on top of each other.
Window faces the other way	Flips the window to the other side of the face.
Show the window open	Slides the bottom sash up (Sash open %), or swings casements and tilts pivot lights (Opening angle) when Flat back is off.
Glass	Adds glass as its own group, so you can hide it or print it in clear. K Glass thickness.

5. The Panes tab

Setting	What it does
Panes across / up	Columns and rows of panes. For a sash these are the **top sash**; the **bottom sash** has its own pair (so 3 x 2 over 3 x 2 is '6 over 6'). For a casement they are each leaf, and the second pair is the top lights.
Bars in the arch	**Fan bars** radiating from the springing, **Grid carries on**, or **Plain glass** above the springing line.
Meeting rail height	Sash only: where the two sashes meet, as a % of the window height.
Casement leaves	How many hinged leaves side by side (1 to 8), split by mullions (J).
Transom height	Casement only: % of the height; 0 = no transom. Above it are top lights.
Pivoting opening light	Fixed only: one pane (column / row, counted from the bottom left) is a separate opening light.

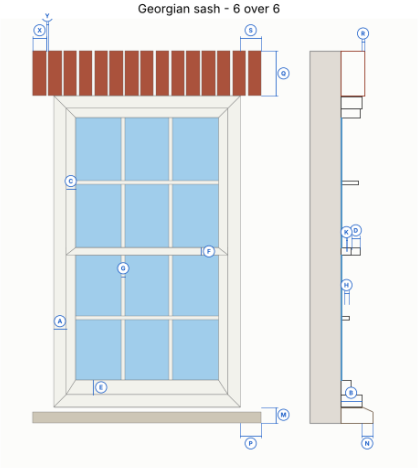
6. The Sill & head tab

Setting	What it does
Sill	**Stone sill** (sloped, M high), **Brick-on-edge sill** (W high, bricks X wide with Y joints) or **No sill**. N = how far it stands out, P = how far it runs past the opening.
Head	**Stone**: a lintel over a square top, a stone arch over a curved top. **Brick**: a soldier course over a square top, a brick arch over a curved top. Q = depth, R = how far it stands out, S = how far a lintel runs past the opening.
Keystone	A tapered keystone in the middle of the head: T wide, rising U above the head and standing V further out.

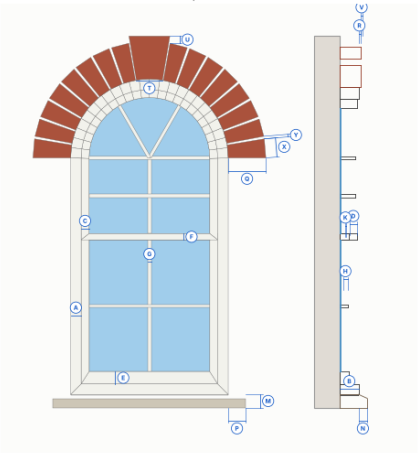
7. Sizes and the lettered drawing

All sizes are full-size millimetres (scale the finished model with 3D Print Scaler). Each size box has a letter, and the **Sizes (A, B, C ...)** tab beside the 3D preview shows a front view and a cut through the window with the same letters, for your current settings. Only the sizes that apply to the chosen style, sill and head are marked.

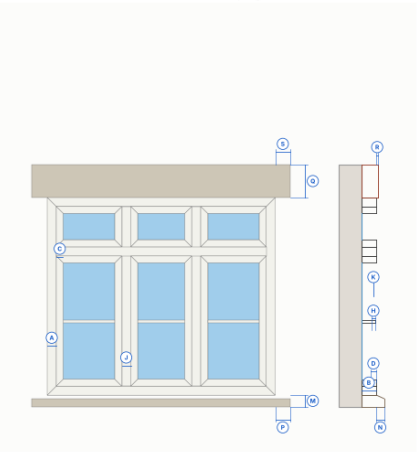
Georgian sash - 6 over 6



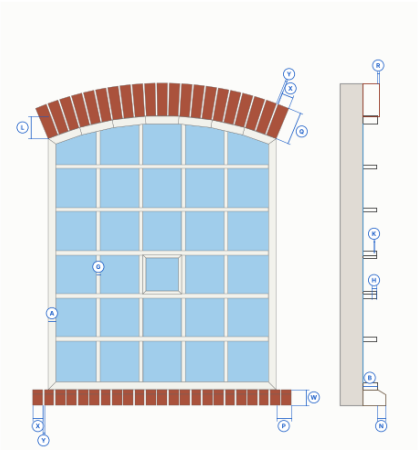
Round-top sash with fan



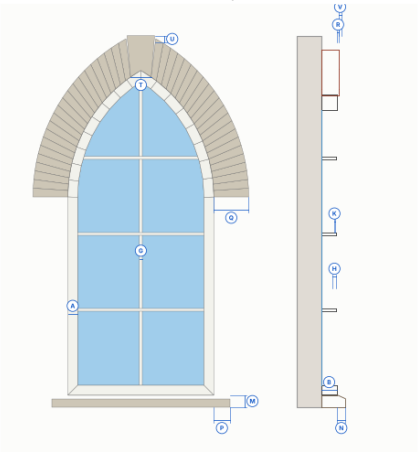
Casement with top lights



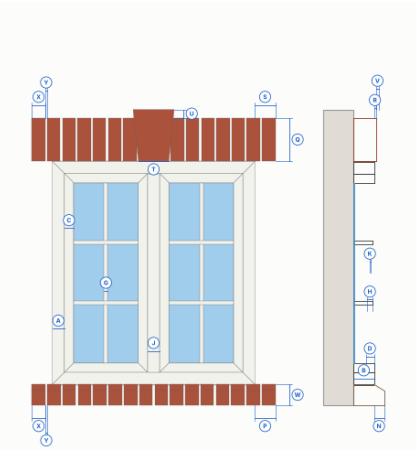
Industrial cast iron - segmental



Gothic chapel



Cottage casement - 2 leaves



	Size		Size		Size
A	Frame width	J	Mullion / transom width	S	Lintel runs past opening
B	Frame depth	K	Glass thickness	T	Keystone width
C	Sash / leaf stiles	L	Arch rise	U	Keystone rises above head
D	Sash / leaf thickness	M	Stone sill height	V	Keystone sticks out more
E	Bottom rail	N	Sill sticks out	W	Sill brick height
F	Meeting rail	P	Sill runs past opening	X	Brick thickness
G	Glazing bar width	Q	Head depth	Y	Mortar joint
H	Glazing bar depth	R	Head sticks out		

8. The 'Last window added' message

After each **Add window** the plugin shows what it did, under the bold line at the top of its window. Green means all went well; orange means something needs a look. You can select the text and copy it.

Message	Meaning
Added 1 group(s), 1314 faces. Join ... on. Cut through: on. Faces replaced: 1.	A summary of what was made.
Cut 1 opening(s) through the wall.	The window, sill and head were cut through to the back of the wall.
Joined into one solid: 1.	The window, sill and head are one watertight solid - ready to export.
No back of the wall was found behind the face ...	The face is a single sheet (or the wall is more than 2 m thick), so nothing was cut; the window sits on the face.
... cut to the window only ...	The wall face round the shape couldn't be found, so only the window shape was cut and the sill and head sit on the wall.
NOT joined ... N of M pieces aren't closed solids ...	IngeTrazo wouldn't join some pieces. The rest are joined and the odd pieces go in a group named '... - pieces not joined'. Send the message to get it fixed.

9. Presets

Pick a preset from the list at the bottom of the window. Built-in presets are complete windows:

Preset	What you get
Victorian sash - 2 over 2	Square sash, 2 panes each sash, stone sill and lintel.
Georgian sash - 6 over 6	Square sash, 3 x 2 panes each sash, stone sill, brick soldier course.
Round-top sash with fan	Semicircular top with fan bars, brick arch and keystone.
Cottage casement - 2 leaves	Two side-hung leaves, 2 x 3 panes, brick sill and head.
Casement with top lights	Three leaves with a transom and top lights, stone sill and lintel.
Industrial cast iron - segmental	Fixed 5 x 6 grid, segmental top, pivot light, brick sill and arch.
Gothic chapel	Pointed arch, fixed 2 x 4 panes, stone sill and arch.
Bull's-eye	Round window with a brick ring and keystone.
Chunky for OO printing (sash)	Thicker frames, sashes and bars so they survive printing at 1:76.

To save your own: set everything up, click **Save...** and give it a name. Each preset is a small .json file in **Documents\IngeTrazo Presets\Windows** (click **Folder** to open it), so you can copy them to another computer. **Delete** removes one of your own presets.

10. Tips for 3D printing

- Leave **Flat back for 3D printing**, **Join into one solid** and **Cut through the wall** on. The wall then prints face up with every window, sill and head flat on the bed.
- Put the windows in **before** the bricks. Then run 3D Brick & Block Studio on the wall's front face with **Closed bricks** ticked and a **Gap round openings** of 0.01-0.05 mm. If PrusaSlicer shows open edges along the bricks at the openings, use 0.05 mm.
- Untick **Soldier course around openings** in the brick plugin when the windows have their own heads and sills - it would follow the combined outline.
- For small scales use the **Chunky for OO printing** preset, or set the bar width (G) to at least about 40-50 mm full size so bars come out at a printable 0.5-0.7 mm at 1:76.
- Add **Glass** if you want to print it separately in clear, or leave it off and glaze with acetate.
- Before exporting, run IngeTrazo's **Solid Inspector**: each window should show as watertight. PrusaSlicer's repair is safe to use on any small errors left in other parts of the model.

11. Troubleshooting

Problem	What to do
The title doesn't say version 9	Close IngeTrazo, copy the newest period_windows.py into the plugins folder over the old one, and restart.
'Select a face first'	Click the face itself (not a group) so it is highlighted, then Add window.
'Not tall enough for a semicircular / pointed top'	The face must be taller than half its width (semicircle) or 0.87 x its width (pointed). Use Segmental, or draw a taller shape.
'The panes don't fit' / 'frame width is too big'	Reduce A (frame width), C (stiles), E or F, or use fewer panes for a small window.
Sill and head on the inside of the wall	Undo, tick 'Window faces the other way', add again.
Preview shows 900 x 1500 mm	Nothing is selected; select the face and click 'Preview selected face'.
Bricks touching the windows in the slicer	Increase 'Gap round openings' in the brick plugin.